Datasets and annotations for different models.

Data	Comment	Link
Full RGB image and thermal/environmental tabular data for each image Of 2024 dataset	In weeks 0,2,3,4 (time points files) folders you can find: ❖ Original RGB images: **Canon** ❖ Cropped RGB images: **Crop_RGB** ❖ Original thermal images: **thermal_thermal_images** ❖ Cropped thermal images: **Crop_thermal_images_tiff** ❖ Segmented RGB images (after U-net): semgnt_img ❖ Extracted meteorological data CSV file for each RGB image (fitted to number of RGB images): **filtered_meteorological_data** ❖ Extracted meteorological data CSV file: **thermal_features** **Notes:** ❖ week 0 excluded from models and week = time point which means it has both days in it (days of time point). ❖ The inner folder of dates are called depending on tree images were taken of.	❖ Time point 1: week-1 ❖ Time point 2: week-2 ❖ Time point 3: week-3 ❖ Time point 4: week-4
Object Detection	In the object detection data, you can find: ❖ PG-YOLO data and annotations. ❖ Data for Transfer learning and annotations. ❖ Output of Raw TL test data images. ❖ test data images. ❖ Sensitivity analysis (TL).	❖ PG-Yolo: PG-YOLO Data ❖ TL: Pom-Obj-Data ❖ Raw TL output: Raw-TL-Output ❖ TL Output: TL-Output
Segmentation	In segmentation data you can find: ❖ Data and masks (after augmentation) ❖ Sensitivity analysis. **Note:** outputs of segmentation you can find in **semgnt_img**	❖ Segmentation data: Pom-Seg-Data
CNN+LSTM model-1 2024 data	In model-1 data you can find: ❖ Initial train and test data for each row.	❖ Train the north side of the field: train-3 ❖ Train the south side of the field: train-4 ❖ Test the north side of the field: test-3 ❖ Test the south side of the field: test-4
CNN+LSTM model-2 2024 data	In model-2 data you can find: ❖ Train and test data for each side of the field (north, south after feature engineering. **Note:** irrigation column exist but not used.	❖ Train the north side of the field: train-3 ❖ Train the south side of the field: train-4 ❖ Test the north side of the field: test-3 ❖ Test the south side of the field: test-4
CNN+LSTM model-3,4,5 2024 data	In model-3, 4, 5 data you can find: ❖ Train and test data for each side of the field after LDA. **Notes:** ❖ image processing is in the code itself locally (not saved as sperate image data) which means it's the same data. ❖ In this model forward all data train and test together.	❖ Train the north side of the field: train-3 ❖ Train the south side of the field: train-4 ❖ Test the north side of the field: test-3

	❖ This data is also used for sensitivity analysis (model-5).	❖ Test the south side of the field: test-4
LDA(CNN+LSTM) model-6 2024 data	In model-6 data you can find: ❖ Train and test data for each side of the field after LDA on CNN image features combined with tabular data.	❖ Train the north side of the field: train-3 ❖ Train the south side of the field: train-4 ❖ Test the north side of the field: test-3 ❖ Test the south side of the field: test-4
CNN+LSTM model-5 Fuzzy Inverse Distance label 2024 data	❖ Train and test data for each side of the field for Fuzzy Inverse Distance label data	❖ Train the north side of the field: train-3 ❖ Train the south side of the field: train-4 ❖ Test the north side of the field: test-3 ❖ Test the south side of the field: test-4
CNN+LSTM model-5 Fuzzy Triangle label 2024 data	❖ Train and test data for each side of the field for Fuzzy Triangle label data	❖ Train the north side of the field: train-3 ❖ Train the south side of the field: train-4 ❖ Test the north side of the field: test-3 ❖ Test the south side of the field: test-4
CNN+LSTM model-5 Fuzzy Gaussian label 2024 data	❖ Train and test data for each side of the field for Fuzzy Gaussian label data	❖ Train the north side of the field: train-3 ❖ Train the south side of the field: train-4 ❖ Test the north side of the field: test-3 ❖ Test the south side of the field: test-4
CNN+LSTM model-5 binary label 2024 data	In model-5 binary label data you can find: ❖ Train and test data for each side of the field after LDA with label 0 representing no cracks and label 1 represent the tree will have cracked fruits.	❖ Train the north side of the field: train-3 ❖ Train the south side of the field: train-4 ❖ Test the north side of the field: test-4 ❖ Test the south side of the field: test-4
Full RGB image and thermal/environmental tabular data for each image Of 2025 dataset	Same structure of 2024 folders.	❖ Time point 1: week-1 ❖ Time point 2: week-2 ❖ Time point 3: week-3
CNN+LSTM model-5 original cracking severity	Structured for model-5	❖ The north side of the field: train-W and test-W ❖ The south side of the field:

labels 2025 data		train-E and test-E
CNN+LSTM model-5 Fuzzy Gaussian label 2025 data	❖ Train and test data for each side of the field for Fuzzy Gaussian label data	❖ The north side of the field: train-W and test-W ❖ The south side of the field: train-E and test-E
CNN+LSTM model-5 binary label 2025 data	In model-5 binary label data you can find: Train and test data for each side of the field after LDA with label 0 representing no cracks and label 1 represent the tree will have cracked fruits.	❖ The north side of the field: train-W and test-W ❖ The south side of the field: train-E and test-E